

Question 1: What is the main difference between supervised learning and unsupervised learning?

Answer: Supervised learning uses labeled training data to learn a mapping from inputs to outputs. Unsupervised learning processes unlabeled data to discover hidden patterns, structures, or groupings within the data without any explicit target label.

Question 2: Explain the concept of overfitting in machine learning models.

Answer: Overfitting occurs when a machine learning model learns the noise and specific details in the training dataset too well, to the extent that it negatively impacts the model's ability to generalize to new, unseen testing data.

Question 3: What is a decision tree and how does it make classifications?

Answer: A decision tree is a supervised model that splits data into subsets based on feature values. It classifies data by passing it down a hierarchy of yes/no decision nodes starting at the root until it reaches a leaf node containing the final class prediction.

Question 4: What is backpropagation and what role does it play in training neural networks?

Answer: Backpropagation is an algorithm used to calculate the gradient of the loss function with respect to the weights of the neural network. This gradient is then used by optimization algorithms like gradient descent to update the network weights and minimize the error.

Question 5: Why are convolutional neural networks (CNNs) particularly well-suited for image processing tasks?

Answer: CNNs use parameter sharing and local receptive fields (using sliding filters) to capture spatial hierarchies and translation invariance in images. This reduces the number of parameters compared to fully connected layers, making them highly efficient and accurate for visual data.

Question 6: Describe the vanishing gradient problem in recurrent neural networks (RNNs).

Answer: In deep RNNs, gradients can decrease exponentially as they are backpropagated through time. This makes it extremely difficult for the model to learn long-term dependencies because the weights associated with early inputs receive virtually no updates.

Question 7: Define the three primary cloud service delivery models: IaaS, PaaS, and SaaS.

Answer: Infrastructure as a Service (IaaS) provides virtualized computing resources like VMs and storage. Platform as a Service (PaaS) provides a platform/environment for developers to build applications without managing OS/hardware. Software as a Service (SaaS) delivers fully functional software applications over the internet.

Question 8: What is virtualization and why is it foundational to cloud computing?

Answer: Virtualization is the technology of creating virtual instances of physical hardware resources. It allows multiple isolated virtual machines (VMs) to run on a single physical host, enabling the resource pooling and scalability needed for cloud environments.

Question 9: What is the purpose of the MapReduce programming model in distributed systems?

Answer: MapReduce is designed to process large-scale datasets in parallel across distributed server clusters. The 'Map' function filters and sorts the data into key-value pairs, while the 'Reduce' function aggregates or summarizes those pairs to produce the final output.

Question 10: What is the role of an activation function, such as ReLU, in a neural network?

Answer: Activation functions introduce non-linearity into the network, allowing it to learn and model complex, non-linear relationships in the data. Without activation functions, a multi-layer neural network would behave exactly like a simple linear regression model.